

Bingchang Wu

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EDUCATION

Georgia Institute of Technology (GPA: 4.0/4.0)

Atlanta, GA

Bachelor of Science in Computer Science; Threads: Intelligence and Systems & Architecture

Expected May 2028

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Organization & Programming, Discrete Mathematics, Linear Algebra, Objects & Design (Software Engineering)

EXPERIENCE

Software Engineering Intern

May 2026 – Aug. 2026

Saite Robotics

- Deployed an internal diagnostic platform that converted ROS logs into technician-ready fault reports, reducing manual fault-triage time by ~80%.
- Built the FastAPI/Streamlit application with PostgreSQL, centralized telemetry, and systemd-managed recovery across VM reboots.
- Integrated RAG into report generation using LangChain, LangGraph, Hugging Face embeddings, and Chroma, grounding reports in maintenance guidance and achieving 93% top-3 retrieval accuracy.
- Developed 6 Python behavior-tree nodes for a separate CARLA simulation task, supporting trajectory evaluation and collision-risk detection while averaging 0.31 m path deviation across 24 scenarios.

Undergraduate Research Assistant

May 2026 – Present

PAIR Lab, Georgia Institute of Technology

- Conducted a controlled visual-navigation study comparing five approaches: single-environment PPO, domain-randomized PPO, behavior cloning with DAGger, zero-shot GPT-4o-mini, and BFS.
- Trained domain-randomized PPO across three layout families, achieving 100%, 98.4%, and 97.8% success across 42, 249, and 276 evaluated layout patterns after 40M steps.
- Built an LLM-as-judge pipeline using a five-category spatial-reasoning taxonomy to classify 1,671 VLM reasoning steps across 40 navigation episodes.
- Increased RL training throughput 8× from ~200 to ~1600 FPS by eliminating redundant observation generation, parallelizing 8 environments, and removing unnecessary GPU overhead.

Section Leader

Apr. 2025 – Jun. 2025

Stanford University, Code in Place

- Selected to teach Python to a cohort of 15 students in Stanford's globally scaled Code in Place course serving 16,000+ learners.
- Led weekly live sections and office hours covering control flow, functions, loops, and data structures through beginner-friendly mental models and hands-on debugging.
- Mentored students one-on-one on problem-solving strategies and project development while adapting pacing for varied experience levels and fostering an inclusive learning environment.

PROJECTS

BuzzBoard | GT Web Dev Club | *Next.js, TypeScript, Express, Firebase, Firestore*

Jan. 2026 – May 2026

- Contributed to a full-stack campus event-discovery platform supporting 500+ campus events.
- Engineered real-time RSVP, friendship, and presence updates using Firestore listeners with optimistic UI updates for instant client-side feedback.
- Designed a proximity-aware feed-ranking algorithm and schedule-conflict detection system, with Leaflet maps and ICS calendar import.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, C, C++, SQL, HTML/CSS

AI/ML: PyTorch, Stable-Baselines3, NumPy, pandas, scikit-learn, MiniGrid, Reinforcement Learning (PPO), Vision-Language Models, Retrieval-Augmented Generation (RAG), Hugging Face, LangChain, LangGraph

Frameworks: Next.js, React, Express, FastAPI, Streamlit, Leaflet

Developer Tools: Git, Docker, Linux (Ubuntu), ROS, Playwright

Databases & Cloud: PostgreSQL, SQLite, Chroma, Firebase, AWS, Vercel, Cloudflare